Low-Contrast Image Enhancement

Histogram equalization: complete explanation with examples

SIMATS Engineering | Computer Vision (ITA05) | Assessment Tool 2: Scenario-Based

1. The scenario and the questions

Scenario: A gray image looks dull. The object and the background have almost the same brightness.

- **(a)** Why does the image look dull?
- **(b)** What problems does it cause?
- **(c)** Which technique can improve it?
- **(d)** Why did you choose this technique?
- **(e)** Write your answer step by step.

How to use this document: Part 2 explains the basic ideas first. Parts 3 to 7 answer (a) to (e) in detail with worked numbers. Part 8 is a short exam-ready answer, and Part 9 has extra viva questions.

2. Basic ideas you must know first

2.1 What is a digital image?

A digital image is a grid of tiny squares called **pixels**. In a gray image every pixel stores one number from **0 to 255**. The number 0 means black, 255 means white, and the numbers in between are shades of gray. These numbers are called **gray levels**. A small gray image of 640 x 480 pixels has 307,200 pixels, and each one carries its own gray level.

2.2 What are brightness and contrast?

Brightness tells how light or dark the image is overall. **Contrast** is the **difference in brightness between the object and the background**. We can see an object only because its gray level is different from the gray level around it. If the object is 30 and the background is 220, the contrast is high and the object stands out. If the object is 118 and the background is 122, the difference is only 4 levels (less than 2% of the full range), so the object almost disappears.

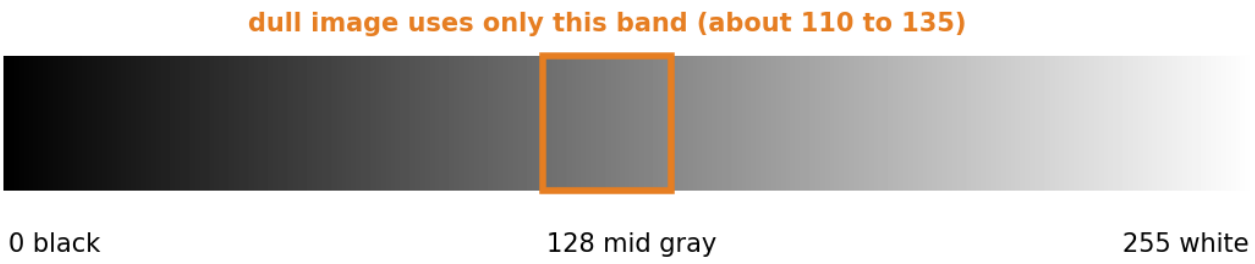


Figure 1: The full gray scale has 256 levels, but a dull image uses only a small band of them.

2.3 What is a histogram?

A **histogram** is a bar graph of the image. The x-axis shows the gray level (0 to 255) and the y-axis shows how many pixels have that level. The shape of the histogram tells us how the image looks.

Histogram shape	What the image looks like
Bunched in a narrow band in the middle	Dull, flat gray, **low contrast** (our scenario)
Bunched on the left (low values)	Too dark
Bunched on the right (high values)	Too bright, washed out
Spread over the whole range 0 to 255	Good contrast, clear details

Key property: in a dull image the object pixels and the background pixels sit very close together in one narrow band of the histogram. The rest of the gray scale (0 to 255) is **not used at all**. This one fact decides which technique we should use.

2.4 What is happening in this scenario, and what do we need?

- The object and the background have almost the same gray level, so the eye cannot separate them.
- We need to **spread the crowded gray levels over the full range** so the small differences become big differences.
- The technique that does this automatically from the histogram is **histogram equalization**.

3. Answer to (a): Why does the image look dull?

Answer: The image has **low contrast**. Almost all pixels have gray levels in a very narrow range, so the object and the background differ by only a few levels. The histogram is bunched together and most of the gray scale is unused.

Cause	What happens	Result in the image
Poor lighting or wrong exposure	Too little or uneven light reaches the camera or scanner, so all the surfaces reflect almost the same amount.	Everything looks medium gray.
Haze, fog or glare	Scattered light is added over the whole scene and hides the real differences.	Washed-out image, object and background merge.
Limited sensor range or faded original	The sensor records only a small part of the brightness range, or an old photo or print has faded.	Narrow histogram, flat look.
Object and background are naturally similar	Both are made of similar material or tone (for example a gray tool on a gray table).	Very small brightness difference between them.

Why can't we see it? The eye notices differences, not absolute values. If the object is 118 and the background is 122, the difference is 4 out of 255 (about 1.6%). That is too small to notice, so the image looks dull and flat.

4. Answer to (b): What problems does it cause?

Answer: 1. Details and edges are hard to see. 2. The object cannot be separated from the background (thresholding and segmentation fail). 3. Later steps such as edge detection, feature extraction and recognition give wrong results.

- **Details are lost.** Small features such as cracks, text, texture or boundaries cannot be seen by the human eye.

- **Segmentation fails.** A threshold separates object from background only if there are two clearly different groups of gray levels. In a dull image there is just one clump, so any threshold cuts the object badly.
- **Edge detection is weak.** Edges are found from the change in gray level between neighbours. If the change is only 4 levels, the edge response is very small and is lost in noise.
- **Recognition and measurement go wrong.** Face detection, medical X-ray analysis, satellite images and quality inspection all depend on clear differences. Wrong boundaries give wrong sizes and wrong decisions.
- **Poor quality for printing and sharing,** even if the information is technically present in the pixels.

5. Answer to (c): Which technique can improve it?

Answer: Use **histogram equalization**. It remaps the gray levels so that the histogram becomes spread over the whole range 0 to 255, which increases the contrast.

5.1 Key words

- **Gray level r:** the value of a pixel (0 to 255). **L** is the number of levels (256 for 8-bit, so L - 1 = 255).
- **Histogram h(r):** the number of pixels that have gray level r. **N** is the total number of pixels.
- **CDF (cumulative distribution):** running total of the histogram. CDF(r) = number of pixels with gray level r **or lower**. It always ends at N.
- **Mapping:** a rule that says which new gray level s each old gray level r becomes.

The equalization formula is:

$s = \text{round}((CDF(r) - CDF_{min}) / (N - CDF_{min}) \times (L - 1))$
where CDF_{min} is the CDF value of the lowest gray level that is present in the image.

5.2 Why does equalization improve a dull image?

In a dull image most pixels are crowded into a few gray levels, so the CDF rises very steeply there. The formula uses the CDF, so the steep part is **stretched over a wide range** of new levels. Levels that were next to each other (for example 118 and 122) are pushed far apart. Levels that nobody uses are skipped. The result is a histogram that is spread over the full range, so the object and the background now have very different brightness.

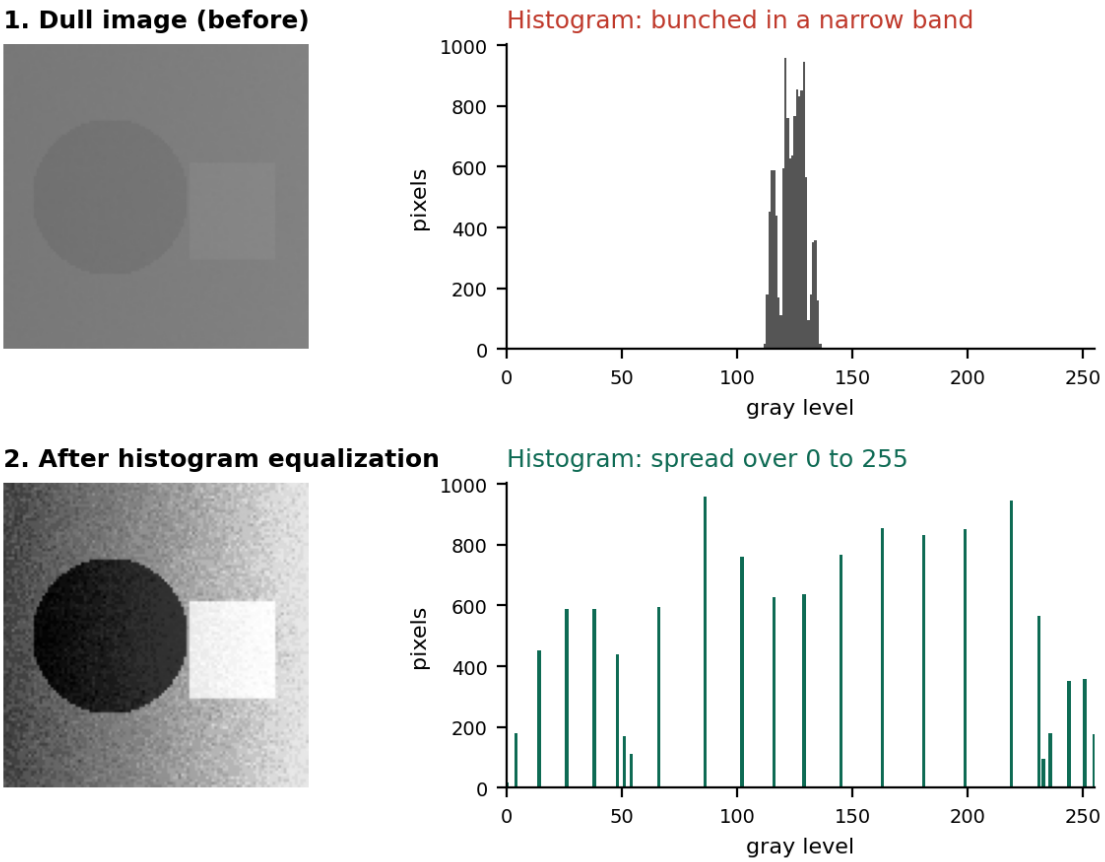


Figure 2: A dull test image (gray levels 111 to 136 only) before and after equalization. The histogram goes from a narrow band to the full 0 to 255 range.

5.3 The steps of histogram equalization

Step	What to do	Why / example
1	Count how many pixels have each gray level (histogram).	Shows how the levels are crowded. Example: level 4 has 5 pixels.
2	Add the counts one after another to get the CDF.	Running total: 3, 8, 13, 16.
3	Apply the formula to every gray level to get its new level s.	Uses CDF, CDF_{min}, N and L - 1.
4	Replace every pixel by its new gray level (look-up table).	Old 4 becomes new 5, and so on, at every position.
5	Check the result.	New histogram is spread over the full range and details are visible.

5.4 Practical points

- **No border problem.** Unlike a filter, equalization works on each pixel by itself using the global histogram, so there is no window and no border pixels.
- **Colour images:** do not equalize R, G and B separately (colours shift). Convert to HSV or YCrCb and equalize only the brightness channel.
- **Noise can also be boosted** because small differences are stretched. For uneven lighting use **adaptive (local) equalization, CLAHE**.

In OpenCV (Python) the same technique is one line:

```
import cv2
img = cv2.imread('dull.png', 0)   # read as gray image
eq  = cv2.equalizeHist(img)       # histogram equalization
cv2.imwrite('enhanced.png', eq)
```

6. Answer to (d): Why did you choose this technique?

Answer: Because the problem is that the gray levels are crowded in a narrow band. Histogram equalization spreads them over the full range **automatically**, needs **no manual settings**, and still works when simpler methods (adding brightness, or linear contrast stretching) fail.

6.1 Comparison with other options

Property	Add brightness	Contrast stretching	Histogram equalization
How it works	Adds a constant to every pixel	Linear stretch of the used range (min to max) to 0 to 255	Remaps levels using the CDF so the histogram spreads out
Does it increase the difference between object and background?	No. Both move together, so the difference stays the same	Yes, but only in proportion to the range	Yes, most strongly where many pixels are crowded
Needs settings from the user?	Yes (how much to add)	Needs min and max	No, fully automatic
Affected by a few extreme pixels?	Can clip bright pixels to 255	Yes, one pixel at 0 and one at 255 stop it working	No, it follows the whole histogram
Best used for	Image that is only too dark	Narrow histogram with a clean min and max	Dull image, crowded or uneven histogram

6.2 Same two pixels, two simple methods

Object = 118, background = 122. For stretching assume the image range is 100 to 140, so s = (r - 100) / 40 x 255.

Method	Object	Background	Difference	Meaning
Original	118	122	4	Almost invisible.
Add 50 (brightness)	168	172	4	Brighter, but contrast is unchanged.
Contrast stretching	115	140	25	Better: difference is about 6 times bigger.

6.3 Where stretching fails but equalization works

Suppose a scan has one black pixel (0) and one white pixel (255), but 99.9% of the pixels lie between 110 and 135. The formula for stretching uses min = 0 and max = 255, so s = (r - 0) / 255 x 255 = r. The image **does not change at all**. Histogram equalization looks at where the pixels really are (the CDF), so it still spreads the 110 to 135 band over the full range. This is why equalization is the safer general choice.

Note for the exam: contrast stretching is a correct simple answer for this scenario too. We choose histogram equalization because it is automatic and more robust, and we mention stretching as the alternative.

7. Answer to (e): Step by step with a full example

Answer: Dull image, find histogram, find CDF, make mapping, replace pixels, enhanced image.

7.1 Example input and output

To keep the numbers small we use a 4 x 4 image with **3-bit** gray levels, so L = 8 and the levels are 0 to 7 (7 = white). The image is dull because it only uses levels 2, 3, 4 and 5. N = 16 pixels.

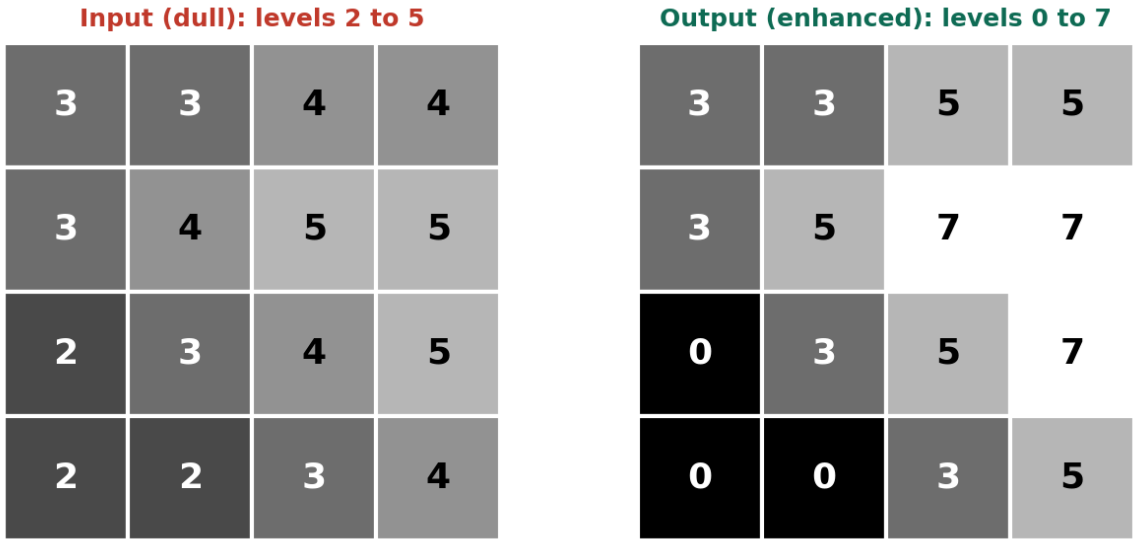


Figure 3: Input (left) and output (right) of histogram equalization. Level 2 became 0 (black) and level 5 became 7 (white).

7.2 Detailed steps

Step 1: Find the histogram (count each gray level)

Level 2 appears 3 times, level 3 appears 5 times, level 4 appears 5 times, level 5 appears 3 times.
Check: 3 + 5 + 5 + 3 = 16 = N. Levels 0, 1, 6, 7 are not used.

Step 2: Find the CDF (running total)

CDF(2) = 3, CDF(3) = 3 + 5 = 8, CDF(4) = 8 + 5 = 13, CDF(5) = 13 + 3 = 16.
CDF_{min} = 3 (the CDF of the lowest level present). The last value 16 equals N, which is a good check.

Step 3: Apply the formula to every level

s = round((CDF(r) - 3) / (16 - 3) x 7) = round((CDF(r) - 3) / 13 x 7)
r = 2: (3 - 3) / 13 x 7 = 0 to s = 0
r = 3: (8 - 3) / 13 x 7 = 2.69 to s = 3
r = 4: (13 - 3) / 13 x 7 = 5.38 to s = 5
r = 5: (16 - 3) / 13 x 7 = 7 to s = 7

Step 4: Replace every pixel using the mapping

2 becomes 0, 3 becomes 3, 4 becomes 5, 5 becomes 7. Apply this at every position to get the output image of Figure 3.

Step 5: Check the result

Before: gray levels 2 to 5 (spread of 3 levels). After: levels 0 to 7 (spread of 7 levels, the full range).

7.3 Table of all calculations

Old level r	Pixels h(r)	CDF(r)	(CDF - 3) / 13 x 7	New level s
2	3	3	0 / 13 x 7 = 0.00	**0**
3	5	8	5 / 13 x 7 = 2.69	**3**
4	5	13	10 / 13 x 7 = 5.38	**5**
5	3	16	13 / 13 x 7 = 7.00	**7**

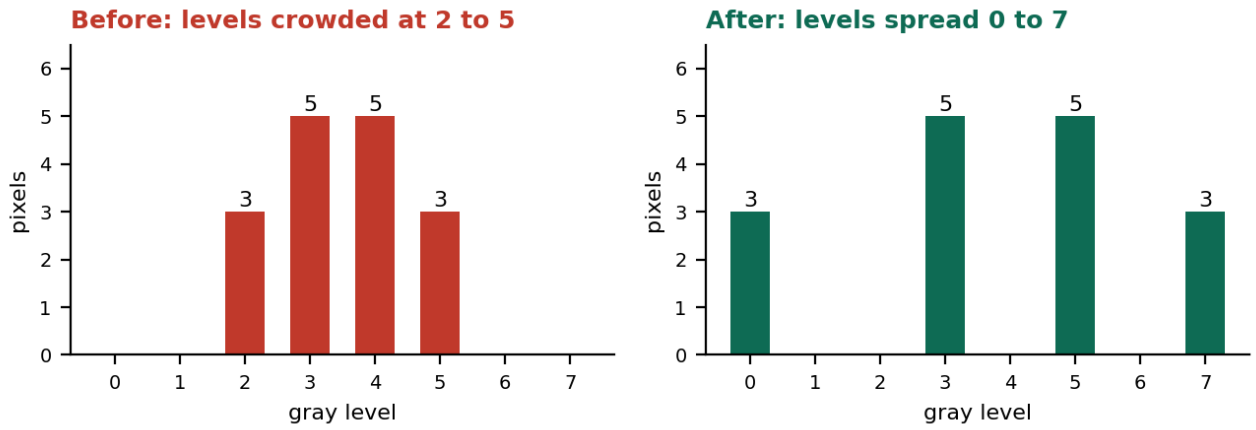


Figure 4: Histogram before and after. The pixel counts stay the same, but the bars move apart over the full range.

7.4 Check your result

- The total number of pixels is still **16** (3 + 5 + 5 + 3). Equalization only moves pixels to new levels, it never adds or removes pixels.
- The darkest level became **0** and the brightest became **7**, so the full range is used.
- The **order** of brightness is kept: 2 < 3 < 4 < 5 became 0 < 3 < 5 < 7. A darker pixel never becomes brighter than a lighter one.
- Levels that were close (4 and 5) are now further apart (5 and 7), so the difference between them is easier to see.

7.5 The complete process in one picture

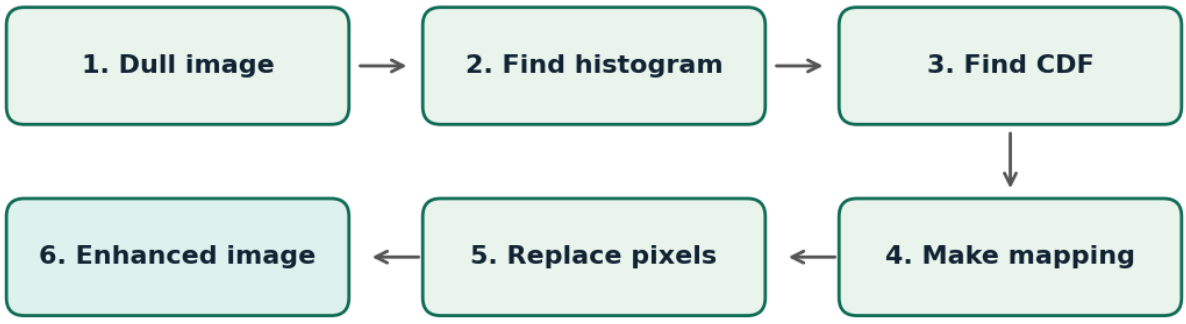


Figure 5: Flow of histogram equalization.

8. Short exam-ready answer

Part	Answer
(a)	The image has **low contrast**. The gray levels of the object and the background are in a very narrow range, so the histogram is bunched together. Causes: poor lighting, haze, limited sensor range, similar object and background.
(b)	1. Details and edges are hard to see. 2. Object cannot be separated from background (segmentation fails). 3. Later steps such as edge detection and recognition give wrong results.
(c)	**Histogram equalization.** Step 1: find the histogram. Step 2: find the CDF. Step 3: map each level with s = round((CDF - CDF_{min}) / (N - CDF_{min}) x (L - 1)). Step 4: replace every pixel with its new level. Step 5: check that the histogram is spread.
(d)	It spreads the crowded gray levels over the full range automatically, needs no manual settings, and works even when a few extreme pixels make contrast stretching fail.
(e)	Dull image, find histogram, find CDF, make mapping, replace pixels, enhanced image.
Example	4 x 4 image, levels 2, 3, 4, 5 with counts 3, 5, 5, 3. CDF = 3, 8, 13, 16. New levels: 2 to 0, 3 to 3, 4 to 5, 5 to 7. The range grows from 2 to 5 up to 0 to 7.
Conclusion	Histogram equalization is the best choice for a dull image because it increases contrast by using the full gray scale.

9. Extra viva questions

Question	Short answer
What is contrast?	The difference in brightness between the object and the background.
What does a narrow histogram mean?	Low contrast. Pixels use only a small part of the gray scale.
What is the CDF?	The running total of the histogram. It tells how many pixels are at or below a gray level.
Is the output histogram exactly flat?	No. Gray levels are whole numbers and a level cannot be split, so it is only approximately flat.
Can equalization be reversed?	Not exactly. Different old levels can map to the same new level, so information can be lost.
Does it change the order of brightness?	No. A darker pixel stays darker than a lighter pixel.
What if the image is already well spread?	Little or no change, because the CDF is already almost a straight line.
What is the problem with global equalization?	It can over-enhance and boost noise, and it ignores local lighting. CLAHE (adaptive equalization) fixes this.
How do you equalize a colour image?	Convert to HSV or YCrCb and equalize only the brightness channel (V or Y).
Difference from contrast stretching?	Stretching is a linear scaling from min and max. Equalization is non-linear and follows the CDF.